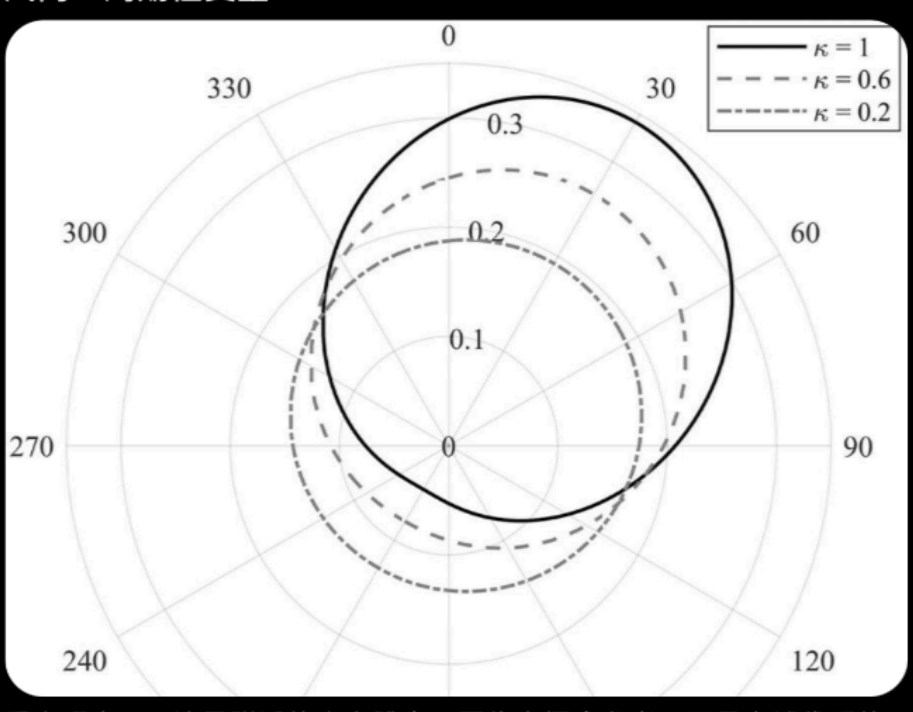
5.10数据生成代码以及绘图

```
import math

import torch

from torch.distributions import VonMises

def wrap_angle(a):

    """将弧度映射到 [-pi, pi)。"""

    return torch.remainder(a + math.pi, 2 * math.pi) - math.pi

def make_wind_data(n=5000, seed=2026, kappa=8.0):

    """

    返回：

      x               [n, 3]  环境特征

      direction       [n]     风向，弧度

      speed           [n]     风速，m/s

      mu_true         [n]     真实条件平均方向

      rate_true       [n]     真实Exponential速率

      mean_speed_true  [n]     真实条件平均风速

    """

    if n <= 0 or kappa <= 0:

        raise ValueError("n and kappa must be positive")

g = torch.Generator(device="cpu").manual_seed(seed)

# 1. 环境输入

    x = 2 * torch.rand(n, 3, generator=g) - 1

    x1, x2, x3 = x.unbind(dim=1)

# 2. 风向的条件中心

    mu = wrap_angle(

        2.8

        + 1.1 * torch.sin(math.pi * x1)

        + 0.65 * x2

        - 0.4 * x3

    )

# 3. 风速的条件均值和速率

    mean_speed = (

        2.5

        + 1.2 * torch.sigmoid(2 * x1 - 0.8 * x2)

        + 0.7 * x3.square()

    )

    rate = 1.0 / mean_speed

# 4. 按von Mises分布采样风向

    # 用fork_rng避免影响外部随机状态

    with torch.random.fork_rng(devices=[]):

        torch.manual_seed(seed + 100_003)

        direction = VonMises(

            mu, torch.full_like(mu, kappa)

        ).sample()

# 5. 按Exponential分布采样风速

    # 逆变换采样：V = -log(U) / lambda

    u = torch.rand(n, generator=g).clamp(

        min=1e-7, max=1-1e-7

    )

    speed = -torch.log(u) / rate

return {

        "x": x,

        "direction": direction,

        "speed": speed,

        "mu_true": mu,

        "rate_true": rate,

        "mean_speed_true": mean_speed,

    }
```

使用示例

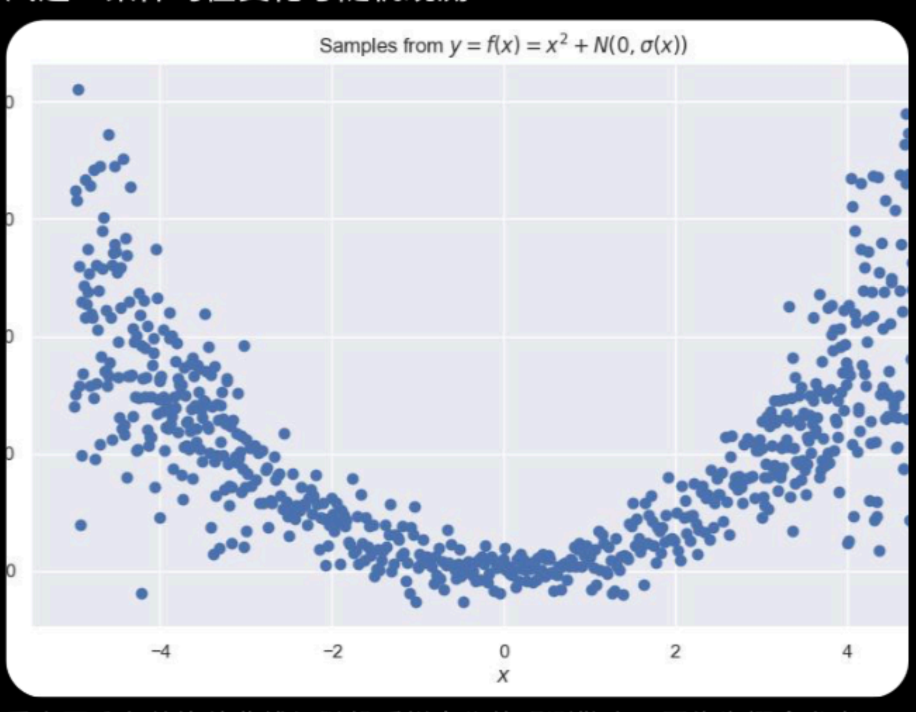
```
train = make_wind_data(n=5000, seed=100)

val = make_wind_data(n=1000, seed=200)

test = make_wind_data(n=2000, seed=300)

print(train["x"].shape)          # [5000, 3]

print(train["direction"].shape)  # [5000]

print(train["speed"].shape)      # [5000]
```

风向：周期性变量



重点观察 ±π 边界附近的方向跳变。图像为概念参考，不是上述代码的实际运行结果。

风速：条件均值变化与随机观测



重点区分条件均值曲线与随机采样产生的观测散点。图像为概念参考。